Prodigy Anesthesia Quick Review

Bier Block

While performing a Bier block, you exsanguinate the arm, inflate the tourniquet to 300 mmHg, and inject 50 mL of 0.5% lidocaine. Three minutes later, the patient complains of a 'pins and needles' sensation in his arm. What should you tell the patient?

On injecting the local anesthetic for a Bier block, the arm discolors and the patient experiences a 'pins and needles' sensation over the next five minutes. You should reassure the patient that this is normal and proceed as planned.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1026.

With what pre-existing conditions is a Bier block contraindicated?

The performance of a Bier block is contraindicated in patient's with severe crush injuries to the arm as hypoxia may worsen the potential outcome, uncontrolled hypertension because it would increase the risk that arterial flow may not be occluded sufficiently, and Raynaud's disease or other forms of peripheral vascular disease affecting the surgical limb. Homozygous sickle cell is also a contraindication to the procedure.

Lee CY. Manual of Anesthesiology. Singapore: McGraw-Hill; 2006: 425-526.

What local anesthetic is most suitable for a Bier block? Least suitable?

Lidocaine 0.5% is suitable for Bier blocks because of the low risk of toxicity at the doses used at a minimum tourniquet time of 20 minutes. Because of its high risk for prolonged cardiotoxicity, bupivacaine is absolutely contraindicated for this procedure.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 970.

What concentration and volume of lidocaine should be used when performing a Bier block?

50 mLs of 0.5% lidocaine

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 970.

Most clinicians limit the use of a Bier block to procedures lasting less than one hour. Why is this?

The pain from the tourniquet becomes intolerable to most patients by about 45 minutes to an hour.

Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1116.

To what pressure should you inflate the tourniquet for a Bier block?

300 torr or 2.5 times the patient's systolic blood pressure.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 970.

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Bier Block

What is the minimum time the tourniquet should be inflated when performing a Bier block? What is the recommended technique for deflating the tourniquet?

To avoid a rapid bolus of intravenous local anesthetic that could result in local anesthetic toxicity, the tourniquet should remain inflated for a minimum of 15 to 20 minutes. Waiting an additional 25-40 minutes as recommended in some sources would prolong the surgical turnover time unnecessarily while deflating the proximal tourniquet and then inflating the distal tourniquet would not be an appropriate action in any situation.

Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1000-1001.

What is the dual-tourniquet technique used in Bier blocks and how is it performed?

Patients can tolerate a procedure under Bier block for a longer period of time if a double-tourniquet is used. The distal cuff is inflated first followed by the proximal cuff. The distal cuff is then deflated to allow the area under it to become anesthetized. If the proximal cuff of the tourniquet becomes too painful for the patient to tolerate, the distal cuff can be inflated and then the proximal cuff deflated to allow the surgeon more time to complete the procedure.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 970.

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